Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/12/2009
PI Name:	Carleton, Stephanie M. Ph.D.		
Project Title:	A Combinatorial Approach of Exercise and Myostatin Inhibition to Enhance Compromised Bone		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	65211-1230	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-02 Postdoctoral Fellowships
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2010 (previously 12/31/2009) per NSBRI (1/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Phillips, Charlotte (MENTOR/University of Missouri-Columbia)		
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	POSTDOCTORAL FELLOWSHIP Myostatin is a member of the transforming growth factor ß (TGF-ß) superfamily and is a negative regulator of skeletal muscle growth. When myostatin is missing or non-functional, the result is uncontrolled muscle growth with a concomitant increase in bone strength. When mice lacking the myostatin gene (mstn/mstn) were subjected to running on a treadmill, their bone strength increased above non-exercised mstn/mstn mice. Exercise has also been shown to be beneficial for human patients at an increased risk for fracture (i.e. osteoporosis, osteogenesis imperfecta). Based on these previous reports, this project was designed to determine if a combination of myostatin inhibition and exercise could improve the bone strength in mice with compromised bone (G610C). The aims of this project are twofold: 1) Determine if a reduction and/or complete absence of the myostatin protein will ameliorate the bone phenotype of heterozygote and		

Task Description:	homozygote G610C mice and 2) Determine if a combination of exercise coupled with reduction and/or absence of myostatin will increase muscle mass and bone strength in heterozygote and homozygote G610C mice. The impact of myostatin inhibition on the geometry and strength of bone has been previously reported. However, a comparison of the gender-specific response of the bone to the loss of the myostatin protein has not been reported and, therefore, was not originally a part of this proposal. However, as the long term goal of this project is to determine if myostatin inhibition is a viable candidate for a pharmacological therapy to improve bone strength, it is imperative to ascertain if a gender difference exists in the response of the bone to myostatin deficiency. Data from male and female mice were analyzed separately and interactions between gender and specific bone parameters were found. This data is the first time a gender-specific response to myostatin deficiency has been reported and adds another dimension to what is known about how the loss of myostatin impacts bone strength. It also necessitates that all future analyses take this gender difference into account and data generated from male mice be analyzed separately from data generated from female mice. Additionally, after the initiation of this project, it was reported that the myostatin genotype of the mother affects the muscle weight of the offspring. Wildtype pups born to mstn/+ mothers had increased muscle weights as compared to those born to wildtype mothers. The same was true for mstn/+ pups. At four months of age, femurs and tibiae from male and female wildtype (Wt), heterozygote (mstn/+) and homozygote (mstn/mstn) myostatin knock-out were removed along with several muscles. Total body mass was found to be greater in mstn/mstn mice as compared to Wt. Males were also larger than females in all genotypes. Muscle mass was larger in mstn/mstn mice than in Wt in both males and females. Gender also impacted muscle mass with males having larger muscles masses than females. Geometry was measured via CT analysis which demonstrated gender differences in all parameters measured, including polar moment of area, with males having longer and wider bones. Additionally, femurs from male mstn/mstn mice had a larger polar moment of area then femurs from either wt or mstn/+ mice. Torsional loading to failure also demonstrated gender differences in whole bone strength and stiffness with males having stronger and stiffer bones than females. Additionally, mstn/mstn males were also significantly stronger and stiffer than male wt or mstn/+ mice. Bone material properties were not different between the genotypes in either male or female mice. Taken together, these data suggest that myostatin inhibition impacts femoral geometry and increases whole bone strength and stiffness without impacting bone material in a gender-specific manner. Due to the complex nature of these analyses with the addition of gender differences and allele status of the mother, we have revised our original breeding scheme. We are currently breeding mstn/+ mice with G610C/+ mice to generate mice of four genotypes: wildtype, mstn/+ mice, G610C/+ mice and mstn/+ G610C/+ mice. Two types of breeding pairs have been set up: 1) mstn/+ mother and G610C/+ father and 2) G610C/+ mother and mstn/+ father. Offspring from these crosses will be divided into two groups, controls and exercisers. Both hindlimb muscles (soleus, plantaris, tibialis anterialis and gastrocnemius) and bones from both groups will be analyzed. Femora and tibiae will be analyzed via CT analysis and torsional loading to failure to determine bone geometry and strength. Muscles will be analyzed for contractile generating capacity, gross pathology and cross-sectional area. Collagen content will also be measured using the hydroxyproline assay in the femora from both groups. Data will be separated based on both gender and maternal myostatin allele status.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Muscle atrophy and bone loss are both been associated with the weightlessness experienced by astronauts, but these are also widespread health concerns on Earth. As a person ages, muscle use tends to decrease. With disuse, muscle atrophy increases and bone strength decreases with a concomitant increase in fracture risk. While it has long been known that exercise decreases muscle atrophy and fracture risk while increasing bone strength, the role that myostatin inhibition may play in further augmenting the positive physiological effects of exercise is not known. This study is designed to determine what impact, if any, myostatin inhibition has on bone strength as well as on muscle physiology and strength in both normal mice and mice with compromised bone whose clinical outcome mimics that seen in osteoporosis patients. Additionally, the effects of exercise on the bone strength in both the normal mice and the mice with compromised bone will also be evaluated to determine if the combination of myostatin inhibition and exercise will prevent both muscle atrophy and bone loss. At the completion of this study we will have determined whether a pharmacological approach of myostatin inhibition should be pursued, along with the inclusion of exercise, to potentially ameliorate the muscle atrophy and bone loss often seen as a consequence of a sedentary lifestyle.
Task Progress:	Toward Aim 1, we have almost completed collecting baseline muscle and bone data on wildtype, heterozygous myostatin knock-out and homozygous myostatin knock-out mice. The impact of myostatin inhibition on the geometry and strength of bone has been previously reported; however, a comparison of the gender-specific response of the bone to the loss of the myostatin protein has not been reported. Data from male and female mice were analyzed separately and interactions between gender and specific bone parameters were found. This data is the first time a gender-specific response to myostatin deficiency has been reported and adds another dimension to what is known about how the loss of myostatin impacts bone strength. It also necessitates that all future analyses take this gender difference into account and data generated from male mice be analyzed separately from data generated from female mice. Additionally, after the initiation of this project, it was reported that the myostatin genotype of the mother affects the muscle weight of the offspring, although the impact on bone strength was not reported. Wildtype and mstn/+ pups born to mstn/+ mothers had increased muscle weights as compared to those born to wildtype mothers. In order to address this, we revised our breeding scheme to generate offspring that will allow us to 1) finish collecting baseline data on the myostatin mice, 2) collect baseline data on the G610C mice and 3) generate the double heterozygote mice (mstn/+ G610C/+) needed to test our hypothesis. The double heterozygote mice will be generated by crossing female mstn/+ with male G610C/+ mice as well as by crossing male mstn/+ with female G610C/+ mice. These crosses will allow us to determine if there is a maternal effect of the myostatin gene on the bone strength of the offspring. We are currently collecting muscles and bone from the offspring of these breeding pairs for analysis.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Abstracts for Journals and Proceedings	Carleton S, Weber B, McCambridge A, Ferriera J, Brown MB, Phillips C. "Impact of exercise on skeletal muscle and bone in oim mice." American Society for Matrix Biology Biennial Meeting 2008, San Diego, CA, December 7-10, 2008. Matrix Biology 2008 Dec;27(Suppl 1):24. , Dec-2008

Abstracts for Journals and Proceedings	Phillips CL, Weber BA, Carleton SM, Mc Cambridge AJ, Ferreira JA, Brown M. "Adult osteogenesis imperfecta model (oim) mice do not exhibit inherent muscle weakness or pathology." 58th Annual Meeting of the American Society of Human Genetics, Philadelphia, PA, Nov. 11-15, 2008. American Society of Human Genetics, November 2008. , Nov-2008
Awards	Carleton SM. "University of Missouri-Columbia Post-Doctoral Association Travel Award, October 2008." Oct-2008